Thank you

for celebrating our student researchers.

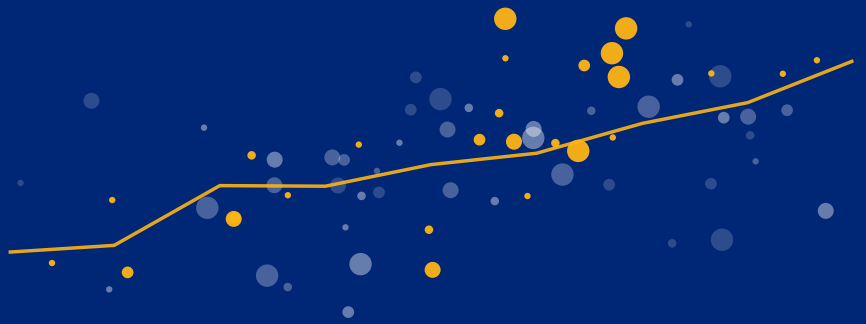
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cdss.berkeley.edu/discovery

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UC Berkeley College of Computing, Data Science, and Society

UC Berkeley College of Computing, Data Science, and Society



CDSS DATA DISCOVERY

Symposium

Spring 2026

Thursday, May 7, 2026

1:30 p.m. – 3:00 p.m.

Lobby of the Hearst Memorial Mining Building

UC Berkeley

PROGRAM

cdss.berkeley.edu/discovery

SPRING 2026 SYMPOSIUM

Welcome

The Data Discovery Program connects teams of Berkeley students with mentors across campus and beyond, using data to understand real-world problems in academia, industry, government and nonprofits.

The Symposium is the culmination of the semester as student researchers share their work through poster presentations. Please join us in celebrating their contributions to the data research community.

You can visit our webpage at https://cdss.berkeley.edu/discovery or email us at cdss-datadiscovery@berkeley.edu.

SCHEDULE

Thursday, May 7, 2026

Lobby of the Hearst Memorial Mining Building, UC Berkeley

1:30 p.m. — **Poster session begins**

2:00 p.m. — **Keynote address**
Assistant Teaching Professor, Dr Josh Grossman

2:15 p.m. — **Presentation of student awards**

3:00 p.m. — **Event concludes**

AT A GLANCE

41	8	5	9
RESEARCH POSTERS	AWARD RECIPIENTS	RESEARCH AREAS	OTHER VENUES

Notes

WITH GRATITUDE

Acknowledgments

We would like to thank everyone who's contributed to the College of Computing, Data Science, and Society's Data Discovery program and making the Symposium come to life. Data Discovery would not be possible without the leadership of Professor Deb Nolan. Associate Dean of Students Professor Narges Norouzi and Robbie Powers have provided invaluable guidance and support. Jessica McDaniels arranged crucial details of the event, and Camille Simsuangco Roxas helped execute it. CDSS Development contributed funds to the Symposium. Claire Chen and Rohan Bhagat have helped keep Data Discovery running smoothly this semester. Ryan Lovett helped arrange access to computing resources and provided technical support, and we acknowledge Research IT and the National Science Foundation's National Artificial Intelligence Research Resource Pilot for providing computational resources (award number NAIRR240387). Finally, we'd like to thank the Data Discovery mentors who provided the projects and mentorship that make the program possible.

CELEBRATING EXCELLENCE

Award Recipients

RESEARCH & STORYTELLING AWARD

Agents in the Loop for Small Molecule Drug Design

Akansha Jain, Annie Gao, Ava Joshi, Gautam Ramkumar, Shreyash Goli

Advised by Alan Cheng, Alec Glisman, Claire Suen, David He and Xiaodong Zhu Merck & Co

RESEARCH & STORYTELLING AWARD

Predicting Total Clay Content from Well Logs

Joseph Wong, Lisa Hao, Theo Dela Cruz

Advised by Kimberly Baldwin & Kevin Smith, U.S. Department of the Interior

RESEARCH & STORYTELLING AWARD

Governing the Sea: Automating Detection of Deep Sea Polymetallic Nodules

Brian Hwang, Kailash Ramesh, Thang Nguyen

Advised by Kimberly Baldwin, Erick Huchzermeyer, Kevin Smith, Obediah Racicot, U.S. Department of the Interior

RESEARCH & STORYTELLING AWARD

Arctic Vegetation Classification Using Machine Learning & Remote Sensing

Jose de Jesus Ochoa Tolento & Thomas Teixeira Scarioli

Advised by Colette Brown, UC Berkeley

RESEARCH & STORYTELLING AWARD

Training and Benchmarking AI/ML Models Using Space Biology Transcriptomic Data

Abraham Guan, Brian Zhou, Karen Meng, Ishanth Hombaiah, Martin Li

Advised by Walter Alvarado, NASA

HONORABLE MENTION

Retinal Laser Scans for Disease Progression Prediction in MS Patients

Adrian Nguyen, Diana Tao, Jenny Shang, Naman Rudrakshi

Advised by Yifan Zhang and Emily Chen, CLight Technologies

HONORABLE MENTION

Identifying Risk Factors of Student Attendance

Edan Wong, Jeremy Li, Olivia Lee

Advised by Ana Costa and Yusuf Ajiboye, Oakland Natives Give Back

HONORABLE MENTION

Can Analysts Predict Markets? Using Equity Research Reports to Forecast Gilead Sciences Stock Price

Amy Tran, Holden Carrillo, Qamil Mirza, Tara Timm, Zhen Liu

Advised by Ethan Yen, Gilead Sciences

41 RESEARCH PROJECTS

Poster Session

Student research teams present their work alongside the mentors and organizations who made it possible.

ENVIRONMENT — 16 posters

Are California School Districts Ready for Climate Impacts?

Karina Parikh, Patrick Chi, Stesha Simon, Zach Makari

Mentored by Devin Ngo & Sarah Whiting, Ten Strands

Analyzing the Landscape of Funding for Environmental & Climate Action

Jason Chang, Natasha Gill, Jeremy Kemmerer, Kelly Le

Mentored by Sarah Whiting, Ten Strands

Operational Decisions and a Changing Climate: Shaping California's Water Future

Aryan Achuthan, Isabelle Wang, Rain Zou, Isabelle Goebel, Sabi Can Ruso

Mentored by Dino Bellugi and James Gilbert, UC Berkeley

Analysis of Surface Albedo and Climate Variability Relationships at Donner Pass

Allen Liu, Christopher Chang, Angelo Payavala, Josh Tsang

Mentored by Gabe Lewis, Central Sierra Snow Laboratory

Arctic Vegetation Classification Via Machine Learning and Remote Sensing

Jose de Jesus Ochoa Tolento, Thomás Teixeira Scarioli

Mentored by Colette Reifsnyder-Brown, Energy and Resources Group, UC Berkeley

Physics-guided AI for Streamflow Forecasting, Flood Control, and Water Supply Prediction in California

Alexander Zhai

Mentored by Dino Bellugi & Sabi Can Russo, UC Berkeley

Exploring Inequitable Access to the SF Bay Shoreline

Ishita Bhadra, Jeffrey Ding

Mentored by Sasha Harris-Lovett, SF Estuary Partnership

Meta-meetings

Amir Rafiei, Peyton Li

Mentored by Dhushan Thevarajah, Coeuraj

Full-Stack RAG Chatbot for Policy Database Search

Bo Wang

Mentored by Karen Chapple & Chenghao Li, Urban Displacement Project, UC Berkeley

Poisson Disk Sampling for 3D Forensic Scene Point Cloud Compression

Arya Prince

Mentored by Jackie Sun, Yu Xuan Bu, Simon Su, NIST

Automated Detection of Accessibility Compliance in Architectural Floorplans Using Computer Vision

Amogh Janganure, Anna Buckeye, Dylan Cha, Noor Rauf

Mentored by Mike Hoppe, Geopogo

CITRIS Health Newsletter and Chatbot Hub

Namira Khanum, Michelle Bao, David Veksler

Mentored by Jolie Lam, CITRIS Health

Training and Benchmarking AI Models for Space Biology

Abraham Westley Guan, Brian Zhou, Ishanth Hombaiah, Karen Meng, Martin Li

Mentored by Walter Alvarado, NASA

PUBLIC POLICY & SOCIAL SCIENCE — 4 posters

NLP for Cuneiform Languages

Zoya Brahimzadeh, Mahi Nagananda, Ojas Sathaye

Mentored by Adam Anderson, Fact Grid

GenAI in the Classroom: Instructor Policy Variation and Shifting Grading Components

Christopher Mach, Lynn Chien, Smrithi Senthilnathan, Sophie McKenna

Mentored by Igor Chirikov, UC Berkeley

HERO2: Incentivizing Sustainable Transportation Behavior through Predictive Analytics & Behavioral Nudging

Aayan Agarwal, Sanjana Alluri, Caden Luu, Aahan Bagga, Carl Djapardi

Mentored by Aurora Garrity, Ryan Xu, Anferney Walther, Alexander Stepanov, HERO2

TokenWorks - OCR for Cuneiform Sources: Evaluating & Refining a Pipeline

Vedashnii Raghu, Tiantong Lu, Rithik Anumula, Jerry Cheng

Mentored by Adam Anderson, Token Works

ALSO PRESENTING ELSEWHERE

These Data Discovery teams are sharing their work at other venues.

TreeAgent: A Multi Agent AI Framework for Tree Labeling and Uncertainty Quantification

Collin Hargreaves, Nicholas Saban, Shiyi Chen

Mentored by Huiqi Wang, UC Berkeley

CANS Scores, Incident Risks, and Formulating Youth Treatment Plans

Brooke Uyeda, Chunmin Zheng, Daniel Lancet

Mentored by Alex Borja, Aspiranet

Analyzing Declining Response Rates in Client Satisfaction Surveys

Matthew Yang, Stephen Tan, Wendy Liu

Mentored by Alex Borja, Aspiranet

Satellite-based Power Plant CO2 Emissions Quantification Using Deep Learning

Ria Voodi, Rohin Juneja, Deleena Ghosh

Mentored by Hikari Murayama and Ronald Cohen, UC Berkeley

Quantifying Power Plant NOx Emissions Using Deep Learning

Pranav Walimbe, Ryuto Tsuruoka

Mentored by Hikari Murayama and Ronald Cohen, UC Berkeley

Meeting the NASEM Grand Challenges: A Data-Driven NLP Temporal Analysis of AEESP Research Publications/Themes

Abhiram Bhavaraju, Ryan Ngo, Riya Sehgal

Mentored by Kara Nelson, Association of Environmental Engineering & Science Professors

Building Data Processing and Analysis Pipeline for Electrochemical Impedance Spectroscopy (EIS)

Anqi Tao, Medha Rakesh, Ayush Guha

Mentored by Raluca O. Scarlat & Matei Ignuta-Ciuncanu, Momoka Imoto, The SALT Research Group

Real World Analysis of Electric Bus Operation in Anaheim Phase II

Annie Liu, Jocelyn Wu, Krithika Muralidhar, Lavin Yau

Mentored by Timothy Lipman, UC Berkeley

Interactive Dashboard Design for Microbial Data Exploration

Martin Sim, Ju Ho Kim, Zouxuan Wu, Jeffrey Gao

Mentored by Jennifer Kuehl & Shekhar Mishra, LBL

How Much Clay? Predicting Mineral Composition from Wireline Logs

Joseph Wong, Theo Dela Cruz, Lisa Hao

Mentored by Kimberly Baldwin, U.S. Department of the Interior

Governing the Sea: Automating Detection of Deep Sea Polymetallic Nodules

Brian Hwang, Thang Nguyen, Kailash Ramesh

Mentored by Kimberly Baldwin, Erick Huchzermeyer, Kevin Smith, Obediah Racicot, U.S. Department of the Interior

AI-Powered Quality Assurance and Quality Control for Water Data

Ciann Amalan, Tanush Obili, Aneya Sobalkar, Joshua Min

Mentored by Dan Wang, CA Water Board

Data Science for a NASA Air Quality Sensor Database

Holden Fees, Ngan Nguyen, Zakaria Al-Alie

Mentored by Kristen Okorn, NASA

2D Spatial Detection and Tracking of Emerging Solar Active Regions Using Acoustic Power Suppression Maps

Umar Ghani, Tanay Pant, Tanish Kher
Mentored by Irina Kitiashvili, NASA

BUSINESS — 6 posters

Optimizing Cross-Platform AI Token Allocation Through Predictive Usage Modeling

Andrew Taylor, Angela Wu, Taewoo Kim, Yirina Wang
Mentored by Nick Cawthon & Ashish Singh, COFAIR

Can Analysts Predict Markets? Using Equity Research Reports to Forecast Gilead Sciences Stock Price

Qamil Mirza bin Abdullah, Holden Carrillo, Zhen Liu, Tara Timm, Amy Tran
Mentored by Ethan Yen, Gilead Sciences

Rivian Product Development AI Assistant

Canhui Huang, Ethan Ngo
Mentored by Zhangli Hu, Rivian

BART: Legacy Codebase Documentation Agent

Rianna Zhu, Haylie Yee, Nikhil Kotta, Carlos Ganoza
Mentored by Yu Shen, BART

LLM-as-a-Judge: Using Large Language Models to Evaluate Machine Translation Quality

Elaine Zhang, Syed Rayyan Ali
Mentored by Goran Muric, InferLink Corporation

Optimal Risk Assessment through Likelihood x Consequence

Carter Chen, Nico Cruz, Reyansh Pallikonda
Mentored by Rodney Martin, NASA Ames Research Center

EDUCATION — 8 posters

Identifying Risk Factors of Student Attendance

Edan Wong, Jeremy Li, Olivia Lee
Mentored by Ana Costa & Yusuf Ajiboye, Oakland Natives Give Back

CourseWise Education Meta-Analytics II: Developing an Internal Metrics Dashboard

Cynthia Wen, Rena Shrestha, Chunmin Zheng
Mentored by Angikaar Singh Chana, Equivalence Systems LLC

Evaluating Small Language Models for AI Education

Ria Jain, Sanjana Sathishkumar, Yiran Hu, SzuLun Huang, Seoha Choi, Lance Santana
Mentored by Eric Van Dusen, UC Berkeley

Does Human Capital Matter?: Analysis Of The Impact Of Dedicated Staff On Environmental And Climate Action In Schools

Pratham Rangwala, Huy Nguyen, Uichan Lee
Mentored by Judy He and Sarah Whiting, Ten Strands

Text Simplification for Children: Evaluating Large Language Models vis-à-vis the human baseline

Cole Fees, Katie Kee, Isole Kim, Tanvi Munjeti, Matthew Takagi
Mentored by Anastasia Smirnova, Experimental and Computational Linguistics Ensemble Lab (ECOLE), San Francisco State University

EmpowerHer Too: Cal Women's Network Mentorship Program

Cadence Loh, Maria Amabella Ava, Rebecca Kong
Mentored by Jolie Lam, CITRIS and the Banatao Institute

Leveraging AI to Assist with Transfer Course Articulation

Erin Kim, Dave Datugan, Shanaya Wickremesinghe, Elizabeth Varghese, Ryan Suh
Mentored by Angikaar Singh Chana, Equivalence Systems, LLC

ConflictQuery: An AI-Powered Compliance Classification System for Academics

Amber Gupta, Jack Hu, Manya Sriram
Mentored by Lawrence Ebringer, ConflictQuery

MEDICINE & PUBLIC HEALTH — 7 posters

Retinal Laser Scans for Disease Progression Prediction in Multiple Sclerosis Patients

Naman Rudrakshi, Adrian Nguyen, Diana Tao, Jenny Shang
Mentored by Yifan Zhang & Emily Chen, C. Light Technologies

Agents in the Loop for Small Molecule Drug Design

Akansha Jain, Ava Joshi, Annie Liao, Gautam Ramkumar, Shreyash Goli
Mentored by Alan Cheng, Alec Glisman, Claire Suen, David He, Xiaodong Zhu, Merck & Co.

LLM-Orchestrated Mixture-of-Experts for Solubility Prediction

Samantha Alonso, Surya Appana, Ameya Kiwalkar, Hilary Wang
Mentored by Alan Cheng & Claire Suen, Merck & Co.

Optimizing DNA Sequencing Pipelines

Anjana Niranjanan, Anthony Ho, Sahana Gopalan, Siya Patel
Mentored by Scott Geller, UC Berkeley DNA Sequencing Facility

The Future of Rapid Research: AI-Powered Preprint Review

Khushi Kolte, Lillian Yao, Noah Fond, Ted Nguyen
Mentored by Angel Paul, Stef Bertozzi, Hildy Fong Baker, RRID